BENCHMARK README

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This repository provides a reproducible framework for benchmarking **USG-TRANSPORT 1.8** under multiple simultaneous model runs ("agents"). The goal is to compare hardware performance across **different CPUs, operating systems, and architectures (ARM / Intel / AMD / etc.)**.

1. Included Scripts

Three automation scripts are provided for both **Windows** and **macOS**:

a. copy_files.* :

- Creates **N copies** of the files directory, each representing a parallel agent.
- Default configuration creates **16 folders**, but users may adjust this.

b. open_and_run.* :

- Launches the run.bat (Windows) or run.sh (macOS) inside each agent folder.
- Default configuration runs **16 agents**.
- To run fewer agents (15→1), adjust the loop end value.

c. clean_up.* :

- Deletes all generated agent folders.
- Default configuration cleans up **16 folders**.

2. Notes for Windows Users

- Scripts are provided as .txt files to avoid antivirus deletion.
- Rename them to:
  - copy_files.bat
  - open_and_run.bat
  - clean_up.bat

3. Notes for macOS Users

- Scripts are provided as .sh files.
- Grant execute permissions before use:
  - chmod +x *.sh

4. Benchmark Engine

- All benchmarks use **USG-TRANSPORT 1.8**.
- The executable is included in this repository.

5. Benchmark Procedure

1. Run copy_files.* to generate agent folders.
2. Run open_and_run.* to start all agents.
3. Benchmark performance for **1 through 16 agents**, adjusting script parameters.
4. After each test, use clean_up.* to delete model files.
5. For each batch of tests, record the **lowest runtime** among all agents.

6. How to Convert and Report Runtimes

- USG-T prints runtimes in this format:

```
Run end date and time (yyyy/mm/dd hh:mm:ss): 2024/08/12 22:10:57
Elapsed run time:  6 Minutes,  6.767 Seconds
```

- Convert this to minutes as:
  - 6 + (6.767 / 60) = 6.1128 minutes
- Round to **4 decimal places**.

7. Recording Your Results in the Spreadsheet

- All contributors must record results in the file:
  - **ByscayneModel_Benchmarks.xlsx**
- The spreadsheet contains **two sheets**:

Sheet 1: Runtimes

Here you will enter:

- CPU Type
- Computer Name
- CPU Cores
- Test ID (e.g., Test1, Test2)
- Runtimes for agents **1 through 16**

CPU Type	i7-10750H	i7-13700KF	i7-14700	i9-9980HK	i9-14900KF	Xeon E-2288G	M1	M2	M2 Pro	M3 Max	M4	M5	Ryzen 7 8845HS	AMD Ryzen 7 2700	Ryzen 9 5950x	Ryzen Threadripper 3990X
Computer	Legion7i	SLG	SA-Model5	Dev	XW Legion T7	DA	SGM001	SGM005	WorkM2	WorkM3	SGM023	WorkM5	Beelink	DA	KOP	ESI
CPU Cores	(12c-2.7GHz)	(16c-5.4GHz)	(20c-5.3 GHz)	(8c-2.3 GHz)	(24c-3.2 MHz)	(8c-3.7GHz)	(8c-3.2 GHz)	(8c-3.49 GHz)	(12c-3.48GHz)	(16c-4.05 GHz)	(10c-4.51 GHz)	(10c-4.61GHz)	(8c-5.1 GHz)	(9 cores 4Ghz, OC)	(16c-4.9 GHz)	(64c-2.9Ghz)
Agent Number	Test1	Test2	Test3	Test4	Test5	Test6	Test7	Test8	Test9	Test10	Test11	Test12	Test13	Test14	Test15	Test16
1	8.17	4.66	8.90	8.12	6.74	7.87	8.32	5.98	6.11	5.10	4.78	4.52	5.30	8.53	5.63	6.68
2	9.52	5.19	9.10	9.38	7.17	8.92	8.73	6.43	6.25	5.29	5.15	4.74	5.35	9.73	6.41	7.00
3	12.60	5.61	9.89	11.07	7.44	10.80	9.27	6.95	6.40	5.55	5.36	5.02	5.88	10.71	7.76	7.17
4	14.82	6.13	11.91	13.97	7.87	13.56	9.47	7.35	6.48	5.67	5.64	5.35	6.67	12.86	9.99	7.32
5	18.00	6.67	12.69	17.58	8.53	15.79	10.78	8.40	6.75	5.82	6.21	5.83	7.62	14.65	12.91	7.38
6	21.23	7.29	15.07	21.00	9.26		11.82	9.23	6.90	5.81	6.76	6.28	9.17	17.47	15.77	7.58
7	24.48	7.95	15.74	24.90	10.15	24.54	13.27	10.47	7.15	5.85	7.33	6.85	10.92	19.90	18.99	7.83
8	27.87	8.91	18.10	27.77	11.38	27.48	14.52	11.33	7.35	6.08	7.89	7.31	12.27	23.73	22.14	8.18
9	31.13	10.21	19.43	31.43	12.62		16.32	12.75	7.81	6.26	8.58	7.85	13.50	27.57	18.99	8.55
10	34.50	11.38	21.34	36.55	13.94		18.13	14.13	8.24	6.75	9.34	8.36	15.00	29.96	21.38	8.95
11	36.72	12.69	24.22	40.86	15.65	37.51	19.60	15.45	8.81	7.43	10.27	9.26	16.50	35.20	25.20	9.38
12	38.28	13.97	29.33	45.93	17.52	41.33	21.43	16.78	9.74	7.90	11.24	10.17	18.17	37.03	29.33	10.05
13	42.80	15.44	32.80	52.18	19.29		23.42	18.15	10.33	8.45	12.20	11.00	20.33	41.27	33.37	10.72
14	45.78	16.64	35.47	54.78	20.65	47.49	24.90	19.73	11.06	9.11	13.18	11.56	22.00	44.78	36.93	11.23
15	52.34	17.81	38.25	59.25	22.58	50.48	26.57	20.90	11.71	9.78	14.10	12.80	24.08	47.97	40.25	11.95
16	54.36	19.08	40.98	61.83	24.10	55.67	28.88	21.98	12.50	10.05	15.06	13.82	27.17	51.21	43.05	12.93

(Use the converted values as described in Section 6.)

Sheet 2: Authors

- This sheet stores metadata about each benchmark run. You must add a new row for each test you contribute.

#	Author	Date	OS	CPU Type	Computer	CPU Cores-Speed	Architechture	Manufacturer	Computer type	Notes
Test1	roh@sgmco.cl	Jan-2023	macOS	i7-10750H	Legion 7i	(12c-2.7GHz)	x86	Intel	Gaming Laptop	
Test2	salvador.lopez.gutierrez@gmail.com	Nov-2025	Win11	i7-13700KF	SLG	(16c-5.4GHz)	x86	Intel	Desktop	
Test3	darce@elmontgomery.cl	Aug-2025	Win11	i7-14700	SA-Model5	(20c-5.3 GHz)	x86	Intel	Desktop	
Test4	roh@sgmco.cl	Jan-2023	macOS	i9-9980HK	Dev	(8c-2.3 GHz)	x86	Intel	MacBook Pro Laptop	
Test5	xiaomin.wang@slrconsulting.com	Oct-2025	Win11	i9-14900KF	Legion T7	(24c-3.2 MHz)	x86	Intel	Gaming Laptop	
Test6	darce@elmontgomery.cl	Aug-2025	Win11	Xeon E-2288G	DA	(8c-3.7GHz)	x86	Intel	Desktop	Incomplete
Test7	roh@sgmco.cl	Jan-2023	macOS	M1	SGM001	(8c-3.2 GHz)	ARM	Apple	Mac Mini	
Test8	roh@sgmco.cl	May-2023	macOS	M2	SGM005	(8c-3.49 GHz)	ARM	Apple	Mac Mini	
Test9	roh@sgmco.cl	Sep-2023	macOS	M2 Pro	WorkM2	(12c-3.48GHz)	ARM	Apple	MacBook Pro Laptop	
Test10	roh@sgmco.cl	Jun-2024	macOS	M3 Max	WorkM3	(16c-4.05 GHz)	ARM	Apple	MacBook Pro Laptop	
Test11	roh@sgmco.cl	Dec-2024	macOS	M4	SGM023	(10c-4.51 GHz)	ARM	Apple	Mac Mini	
Test12	roh@sgmco.cl	Nov-2025	macOS	M5	WorkM5	(10c—4.61GHz)	ARM	Apple	MacBook Pro Laptop	
Test13	jrumbaugh@groundwatermodels.com	Dec-2024	Win11	Ryzen 7 8845HS	Beelink	(8c-5.1 GHz)	x86	AMD	Deskop Mini Box	
Test14	darce@elmontgomery.cl	Aug-2024	Win11	Ryzen 7 2700	DA	(9 cores 4Ghz, OC)	x86	AMD	CPU overclocked by user	
Test15	fakoppli@uc.cl	Jun-2024	Win11	Ryzen 9 5950x	KOP	(16c-4.9 GHz)	x86	AMD	Gaming Desktop	
Test16	jrumbaugh@groundwatermodels.com	Oct-2023	Win11	Ryzen Threadripper 3990X	ESI	(64c-2.9Ghz)	x86	AMD	Desktop	

8. Contributing to the Dataset

- Please submit your benchmark results via: **Pull request** or via **Email** (Pull request is preferred).
- Your contributions will help build a **community-supported benchmark dataset** to compare groundwater model performance across CPU architectures and operating systems.